Curriculum Vitae

Personal information



Name: Chihang He

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Education

- **Huazhong University of Science and Technology** **2023–Present**
  Advisor: Prof. Chao Liu
  Thesis: Proofs of singularity-free solutions and scalarization in nonlinear Einstein-scalar-Gauss-Bonnet cosmology
- **Huazhong University of Science and Technology** **2019–2023**
  B.S. in Physics
  Relevant Courses: General Relativity, Quantum Field Theory, Differential Geometry

Research Experience

- **Holographic Mapping of Particle Dynamics** **2022–2023**
  Advisor: Prof. Wuzhong Guo, HUST
  - Established operator-state correspondence for AdS geodesics in CFT.
  - Proved semiclassical consistency ($G \to 0$) between commutators and Poisson brackets.
  - Derived conformal transformation rules for initial conditions.
- **Singularity-Free Cosmology in ESGB Gravity** **2023–Present**
  Advisors: Prof. Chao Liu, HUST, Prof. Jinhua Wang, XMU
  - Class. Quantum Grav. 43 125005, arXiv:2512.00455
  - First rigorous proof of global singularity-free solutions in nonlinear ESGB theory.
  - Developed the *power identity* method to derive structural differential inequalities.

Publications

- **He, C.**, Liu, C., Wang, J. (2025). "Proofs of singularity-free solutions and scalarization in nonlinear Einstein-scalar-Gauss-Bonnet cosmology"
  Class. Quantum Grav. 43 125005
- **He, C.**, Liu, C.(2025). "Existence and bounds of nonlinear singularity-free cosmological solutions in a string-inspired gravity"
  arXiv preprint arXiv:2512.00455
- **He, C.**, Liu, C.(2026). "Kasner critical scaling in Einstein–Maxwell–scalar black holes with nonlinear collapse and scalarization"
  arXiv preprint arXiv:2609.35500